Supplementary Information: 700-km intercity Yb/Sr optical clock comparison at 2.3×10^{-18}

xxxxx

1 Per-segment data

Table 1: Per-segment data of the 16 segments analyzed in this work. Ratio deviations $y_i = R_i/R_1 - 1$ are relative to the first segment, with $R_1 = 1.207\,507\,039\,343\,337\,721\,0$. $u(y_i)$ is the statistical uncertainty of the segment ratio; "Fit" is *est.* for segments whose stability fit lies in the white-frequency-noise band and *lim.* for model-limited fits.

Segment	n_{valid}	y_i (10^{-18})	$u(y_i)$ (10^{-18})	Fit
1	68 009	+0.00	4.245	est.
2	29 481	-3.36	3.017	lim.
3	42 891	+2.00	5.880	est.
4	64 280	-4.61	3.674	est.
5	19 532	-3.38	15.250	est.
6	52 140	-1.15	2.230	lim.
7	39 526	+5.31	3.476	lim.
8	43 565	+0.97	9.860	est.
9	57 938	-0.76	5.003	est.
10	35 824	+2.61	5.976	est.
11	40 346	+0.73	4.843	lim.
12	152 994	-3.09	4.019	est.
13	68 618	+1.97	6.254	est.
14	57 649	-0.25	4.028	est.
15	147 640	-0.77	1.809	lim.
16	65 983	-0.89	8.813	est.

2 Within-segment tidal correlation analysis

3 Frequency-chain relations and the ratio inversion

The two clock transitions are bridged to the 1 550 nm telecom carrier through fixed factor-of-two relations. On the Shanghai side the Sr 698 nm transition is doubled to 1 397 nm; on the Wuhan side the Yb 578 nm transition is doubled to 1 156 nm. Both ends are phase-locked to independent optical frequency combs whose repetition rate is $f_{\mathrm{rep}} = 200$ MHz, derived from a 10 MHz reference divided by 20.

The recorded observable is the out-of-loop beat b between the received carrier and the local Yb-referenced comb, sampled every second. The ratio inversion uses the fixed chain constants

Table 2: Leave-one-out sensitivity of the ratio combination. Each row excludes one of the 16 segments and recomputes the combination from the remaining segments. The center values are stable at the 10^{-18} level under any single exclusion.

Excluded segment	R_{wls}	χ^2_{red}	u_{wls} (10^{-18})
1	1.207 507 039 343 337 720 4	0.542	1.136
2	1.207 507 039 343 337 720 8	0.474	1.164
3	1.207 507 039 343 337 720 3	0.531	1.123
4	1.207 507 039 343 337 720 8	0.451	1.145
5	1.207 507 039 343 337 720 4	0.540	1.111
6	1.207 507 039 343 337 720 6	0.535	1.217
7	1.207 507 039 343 337 719 9	0.347	1.150
8	1.207 507 039 343 337 720 4	0.542	1.114
9	1.207 507 039 343 337 720 4	0.543	1.128
10	1.207 507 039 343 337 720 3	0.525	1.122
11	1.207 507 039 343 337 720 4	0.539	1.130
12	1.207 507 039 343 337 720 6	0.512	1.139
13	1.207 507 039 343 337 720 4	0.533	1.121
14	1.207 507 039 343 337 720 4	0.543	1.139
15	1.207 507 039 343 337 720 6	0.540	1.287
16	1.207 507 039 343 337 720 4	0.543	1.115

Constant	Value	Meaning
N_{1156}	1,295,739	Yb 1156 nm comb count
N_{1397}	858,456	Sr 1397 nm comb count
N_{1550}	773,598	1550 nm link count (Shanghai end)
N_{1550}^{WH}	966,996	1550 nm link count (Wuhan end)
f_{ref}	10 MHz	comb repetition base
div_{20}	20	divider ($f_{\mathrm{rep}} = 200$ MHz)

The beat is converted to fractional units with the transfer frequency $f_{1550} = 193.3992\,\mathrm{THz}$ $(= N_{1550}^{\mathrm{WH}}\, f_{\mathrm{rep}})$. The segment-mean beat deviation $\overline{\delta b}$ enters the Sr/Yb ratio through

$$\mathrm{Dr} = \frac{c_{1156}}{N_{1156}} \frac{\overline{\delta b}}{f_{\mathrm{ref}}\,\mathrm{div}_{20}} \Bigg/ \left[\frac{c_{1397}}{N_{1397}}\left(N_{1550} + \frac{7}{25} + \frac{1}{25}\right)\right],$$

where $c_{1156} = (1 + b_{\mathrm{Yb}})/2$ with $b_{\mathrm{Yb}} = -1.7 \times 10^{-18}$ (Yb-side fixed shift) and $c_{1397} = (1 + a)/2$, with a collecting the five Sr systematic components listed below. The segment ratio is

$$\mathrm{Sr/Yb} = \mathrm{ratio}_{\mathrm{base}} + \mathrm{Dr}, \qquad \mathrm{ratio}_{\mathrm{base}} = \frac{c_{1156}}{N_{1156}} \frac{N_{1550}^{\mathrm{WH}} + \frac{26}{20} + \frac{m}{f_{\mathrm{ref}}\,\mathrm{div}_{20}}}{\frac{c_{1397}}{N_{1397}}\left(N_{1550} + \frac{8}{25}\right)},$$

where m is the campaign-global median of the plausible beat samples. The static gravitational-redshift correction is applied as a multiplicative factor,

$$(\mathrm{Yb/Sr})_{\mathrm{corr}} = (\mathrm{Yb/Sr})_{\mathrm{raw}}\,(1 + \delta_g), \qquad \delta_g = -3.116 \times 10^{-15}.$$

All ratio arithmetic is carried out in 80-digit decimal precision, mirroring the campaign processing convention.

The Sr systematic shift a is the sum of five components evaluated per segment: lattice AC-Stark (a_{rou}, about -2.3 to -3.5×10^{-19}), AC Stark (a_{AC}, $+7.44$ or $+8.93 \times 10^{-18}$ by campaign period), second-order Zeeman (a_{SM}, the dominant term), air refractive index ($a_{\mathrm{air}} =$

-6.7×10^{-19}), and blackbody radiation ($a_{\mathrm{BBR}} = 0$). The first three are segment-dependent constants recorded in the campaign parameter file; the last two are fixed. Full per-segment values are reproduced in the frozen result layer referenced in Data availability.

4 Statistical model and combination

Each segment i contributes a ratio estimate R_i and a statistical uncertainty u_i obtained from the overlapping Allan deviation $\sigma_y(\tau)$ of the beat fractional frequency, evaluated on a tau grid of powers of two plus 600, 1200, 3600 and 7200 s, limited to a quarter of the longest gap-free run of the segment. u_i adopts the last computable point, $u_i = R_i\,\sigma_y(\tau_{\max})$, without extrapolating the stability curve beyond the data; the log-log slope of $\sigma_y(\tau)$ is retained as a white-frequency-noise diagnostic only.

The choice of the last computable point is deliberate and conservative. The beat record develops a low-frequency plateau for $\tau \gtrsim 4096\,\mathrm{s}$ (a flicker-frequency-noise floor from the clocks and the link, not the tides); fitting the white-frequency slope and extrapolating it to the full segment duration would systematically understate u_i by 30–40 %, inflating the reduced chi-square to about 4 and making the segments appear mutually inconsistent. Adopting σ_y at the largest computable τ instead keeps the statistical uncertainty honest at the expense of being somewhat conservative: the resulting $\chi^2_{\mathrm{red}} = 0.41$ and Birge ratio 0.64 are below unity, indicating that the segments are consistent with their (slightly conservative) uncertainties. Segments whose log-log slope leaves the white-frequency-noise band $[-0.65, -0.35]$ are flagged as model-limited but retained with their measured uncertainties; they lie at the steep end of a continuous slope distribution with no common campaign, weather or laboratory condition.

The segments are combined under a random-effect model with common excess scatter, $R_i \sim \mathcal{N}(\mu, u_i^2 + \xi^2)$. Three estimators are evaluated:

- *Weighted least squares* with inverse-variance weights $w_i = 1/u_i^2$: $\hat{\mu} = \sum w_i R_i / \sum w_i$, with $u_{\mathrm{wls}} = 1/\sqrt{\sum w_i}$. The reduced chi-square $\chi^2_{\mathrm{red}} = \sum (R_i - \hat{\mu})^2/u_i^2/(N-1)$ and the Birge ratio $\sqrt{\chi^2_{\mathrm{red}}}$ quantify the segment agreement.
- *Mandel–Paule*: the same observation model with ξ chosen so that $\chi^2_{\mathrm{red}}(\xi) = 1$.
- *Bayesian random effect*: flat priors on μ and on $\log \xi$, with the joint posterior marginalized numerically over a 40-decade $\log \xi$ grid (401 points, trapezoidal quadrature). The grid span is expanded until the posterior mass at the edges falls below 1 %. The posterior mean of μ and its posterior standard deviation give the reported central value and statistical uncertainty.

All three center values are reconstructed in decimal precision from the segment-1 baseline, $R = R_1\,(1 + \hat{\mu}/R_1)$, so that differences at the 10^{-18} level are not lost in floating-point arithmetic.

5 Leveling details and national-standard comparison

The first-order spirit leveling line runs about 1 113 km between the Wuhan benchmark WS-03 and the Shanghai benchmark WS-01 through Ezhou, Huangshi, Huanggang, Lu'an, Hefei, Wuhu, Guangde and Suzhou. Digital levels with bar-code invar staffs were used, with sight lengths not exceeding 30 m; the line was observed in 6 km to 8 km sections in both directions, each checked in the field, and gravity was observed along the line at the leveling points with two relative gravimeters. The per-kilometer accidental error is $M_\Delta = 0.255\,\mathrm{mm\,km^{-1}}$. The most stringent first-order requirements of the national standards of China (GB/T 12897–2006, $M_\Delta \le 0.45\,\mathrm{mm\,km^{-1}}$), the United States (FGCC 1984, $0.5\,\mathrm{mm\,km^{-1/2}}$), Germany (AdV FA-PN 2022, 0.4 mm) and Czechia (ČÚZK 2015, $m_0 = 0.40\,\mathrm{mm\,km^{-1}}$) are all satisfied.

The independent GNSS-plus-geoid check used precise point positioning for the ellipsoidal heights of the permanent marks, local leveling for the vertical offsets to the clock reference surfaces, and the EGM2008 geoid model for the conversion to geopotential, giving $280.60 \pm 2.558\ \mathrm{m^2\,s^{-2}}$, consistent with the leveled value but about 30 times less precise, so it serves as a consistency check rather than an independent calibration at the reported accuracy.

6 Tidal template construction and the two tide models

The primary tide template is the solid Earth tide and ocean tide loading displacement difference between the two sites, computed every 30 s with the public software ETideLoad4.5, which implements the geophysical models and numerical standards of the IERS Conventions (2010). The displacement difference is converted to an equivalent geopotential difference with the local gravity $g = 9.794\,\mathrm{m\,s^{-2}}$ and then to a beat-frequency template $h = f_{1550}\,\Delta W/c^2$ with the same f_{1550} normalization as the beat record; the direction convention is $\Delta W = W(\mathrm{Wuhan}) - W(\mathrm{Shanghai})$.

The cross-check template computes the solid tide from the DE421 ephemerides (Earth–Sun–Moon positions) with Love numbers, and takes the ocean-tide loading correction to the gravitational potential directly from the FES2022 global ocean-tide model, on a 90 s Beijing-time grid. The two templates are independent realizations; their mutual correlation over the campaign is +0.88, and their segment-mean predictions differ in detail (Figure 1b).

The historical response amplitude $A = +0.54$ used for the fixed-amplitude combinations comes from the demeaned beat waveform of an earlier analysis of the same campaign: the precision-weighted amplitude of a 1200-s-windowed, demeaned fit of the beat against the ETideLoad4.5 template was +0.54, with a standard uncertainty of about 0.08. As an independent consistency check, a grid scan of fixed amplitudes $A_{\mathrm{fix}} \in \{-1.0, -0.54, -0.30, -0.20, 0, +0.20, +0.54, +1.0\}$, applied to the within-segment residuals, finds its residual-correlation zero crossing at $A_{\mathrm{fix}} \approx +0.54$, matching the fitted value; the same analysis locates the dominant coupling in the ocean-tide loading term, since the solid tide alone gives a markedly weaker segment-mean correlation (see the main text). The amplitude is carried as a fixed historical value rather than a free parameter of this work.

7 Transfer subsystems

The fiber route is divided into spans of tens of kilometers, each terminated by a relay station carrying a narrow-linewidth laser, an erbium-doped fiber amplifier and a fiber-noise-cancellation loop; 21 independent relay stations carry 22 amplifiers in total. The link extends the field-deployed cascade of a previous 2 067 km demonstration, and its added instability is kept below the clock noise for averaging times beyond 100 s. Both frequency combs are self-referenced and phase-locked to their local clock signal; all other frequencies in the chain derive from the two clock transitions through the fixed factor-of-two relations of Section 3, so the two clocks are the system's only independent frequency sources. The fiber loop-back is monitored out of loop throughout. The frequency-conversion stages each contribute 0.1×10^{-18} and 0.1×10^{-18} to the uncertainty budget from their frequency-reference settings.

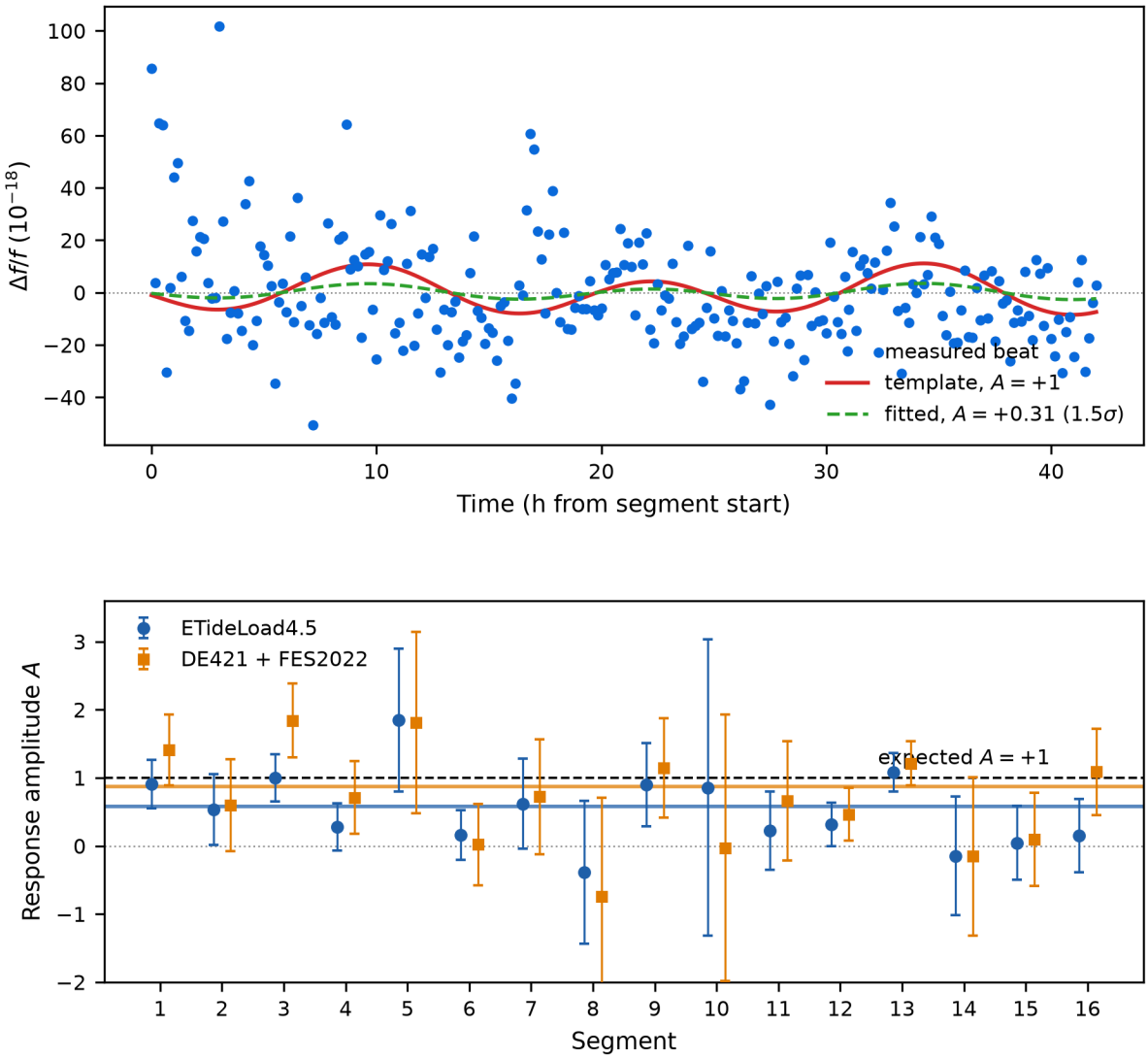


Figure 1: Within-segment tidal correlation analysis. **(a)** Single-segment example: the 1200-s demeaned beat waveform (dots) and the tidal template scaled at the expected response $A = +1$ and at the fitted amplitude, for the ETideLoad4.5 model. No single segment reaches a significant correlation on its own. **(b)** Per-segment response amplitudes A for the two tide models, over the 16 segments entering the ratio combination. Error bars are per-segment fit uncertainties from non-overlapping windows; the solid lines mark the precision-weighted combinations. In 12 of the 16 segments the correlation is positive for both models, and both precision-weighted amplitudes (+0.58 and +0.87) fall short of the expected response of +1 (dashed line), so the observed tidal coupling is smaller than any single template predicts.